Continuous Time Financial Economics

Homework 1

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Question	Grade	/Max
Q1	10	/10
Q2	9	/10
Q3	10	/10
Q4	8	/10
Q5	10	/10
Total	47	**/50**

Comments:

Well done!

***** Please refer to the course slides for instructions.**

Q1. Numeraires and Probabilities

- We wish to understand how the distribution of a stock (that is, its probability measure) changes as we change the numeraire.
- In this question, we wish to consider as numeraire an *arbitrary risky asset (denoted Y) that does not pay dividends.* (In class, we already studied two numeraires: (1) the risk-free bank deposit, and (2) a dividend-paying stock; the classnotes will help you solve this problem.)
- Let $r(t)$ denote the instantaneous risk-free rate at date t, and let $R(t) = \exp(\int_0^t r(u)du)$, so that:

$$\frac{dR}{R} = r(t)\,dt.$$

- Let S be the price of an asset that has a constant dividend yield q. Let $V(t) = e^{q\,t}S(t)$ be the *value* of the portfolio in which all dividends are reinvested, so that:

$$\frac{dV}{V} = \underbrace{q\,dt}_{\text{dividends}} + \underbrace{\frac{dS}{S}}_{\text{capital gains}},$$

with

$$\frac{dS}{S} = \mu_s\,dt + \sigma_s\,dB_s.$$

- Finally, let Y be the price of another asset that does not pay any dividends:

$$\frac{dY}{Y} = \mu_y\,dt + \sigma_y\,dB_y,$$

where B_s and B_y are Brownian motions under the physical (actual) probability measure with correlation ρ, and $\{\mu_s, \mu_y, \sigma_s, \sigma_y,\ \rho\}$ are general random processes.

Please answer the following questions.

1.1 Show that if Y is the numeraire, then the drift of the process dY/Y equals $r + \sigma_y^2$. *Hint:* To do this, identify the drift of the process dC/C where $C(t) = R(t)/Y(t)$.

1.2 Now, using the result in the previous part, identify the drift of dS/S by finding the drift of dC/C, where $C(t) = V(t)/Y(t)$.

1.3 Finally, show that the results derived in class are special cases of the result derived above. That is, show that if Y is the price of instantaneously risk-free asset so that $\sigma_y = 0$, then your answer to the previous question implies that the drift of dS/S is $r - q$. And, if Y is the price of the risky stock S, then the drift of dS/S equals $r - q + \sigma_s^2$.

Answer to Q1 starts here.

Q1.1

Define the process $C(t) \equiv R(t)/Y(t)$. Using Itô's formula for ratios:

$$\begin{aligned}
\frac{dC}{C} &= \frac{dR}{R} - \frac{dY}{Y} + \left(\frac{dY}{Y}\right)^2 \\
&= r\,dt - \left[\mu_y\,dt + \sigma_y\,dB_y^{\mathbb{P}}\right] + \sigma_y^2\,dt \\
&= -\left[\mu_y - r - \sigma_y^2\right]dt - \sigma_y dB_y^{\mathbb{P}} \\
&= -\sigma_y\left[\frac{\mu_y - r - \sigma_y^2}{\sigma_y}\,dt + dB^{\mathbb{P}}\right] \\
&= -\sigma_y dB^{\mathbb{Y}}
\end{aligned}$$

where we define the Itô process $B^{\mathbb{Y}}$ by $B_0^{\mathbb{Y}} = 0$ and

$$dB^{\mathbb{Y}} = dB^{\mathbb{P}} + \frac{\mu_y - r - \sigma_y^2}{\sigma_y}\,dt.$$

Or equivalently:

$$dB^{\mathbb{P}} = dB^{\mathbb{Y}} - \frac{\mu_y - r - \sigma_y^2}{\sigma_y}\,dt.$$

*T*he drift adjustment from $B^{\mathbb{P}}$ to $B^{\mathbb{Y}}$ corresponds precisely to the measure change described by Girsanov's theorem, ensuring that the process $C(t)$ (under the new measure $\mathbb{Y}$) is a martingale. Substituting into the process dY/Y, we obtain:

$$\begin{aligned}
\frac{dY}{Y} &= \mu_y\,dt + \sigma_y\,dB_y^{\mathbb{P}} \\
&= \mu_y\,dt + \sigma_y\left(dB^{\mathbb{Y}} - \frac{\mu_y - r - \sigma_y^2}{\sigma_y}\,dt\right) \\
&= \mu_y\,dt + \sigma_y dB_y^{\mathbb{Y}} - \mu_y\,dt + r\,dt + \sigma^2\,dt \\
&= (r + \sigma_y^2)\,dt + \sigma_y dB^{\mathbb{Y}}
\end{aligned}$$

Then we proved that the drift of the process dY/Y is $(r + \sigma_y^2)$

Q1.2

Let $C(t) = \frac{V(t)}{Y(t)} = \frac{e^{qt}S(t)}{Y(t)}$. We are given:

$$\frac{dV}{V} = q\,dt + \frac{dS}{S}, \quad \frac{dS}{S} = \mu_s\,dt + \sigma_s\,dB_s, \quad \frac{dY}{Y} = \mu_y\,dt + \sigma_y\,dB_y, \quad dB_s \cdot dB_y = \rho\,dt$$

Apply Itô's formula to the ratio $C(t) = \frac{V(t)}{Y(t)}$:

$$\begin{aligned}
\frac{dC}{C} &= \frac{dV}{V} - \frac{dY}{Y} + \left(\frac{dY}{Y}\right)^2 - \left(\frac{dV}{V} \cdot \frac{dY}{Y}\right) \\
&= (q\,dt + \mu_s\,dt + \sigma_s\,dB_s) - (\mu_y\,dt + \sigma_y\,dB_y) + \sigma_y^2\,dt - \sigma_s\sigma_y\rho\,dt \\
&= \big(q + \mu_s - \mu_y + \sigma_y^2 - \sigma_s\sigma_y\rho\big)\,dt + \sigma_s\,dB_s - \sigma_y\,dB_y
\end{aligned}$$

Now, under the measure associated with the numéraire $Y(t)$, the process $C(t)$ must be a martingale. This implies that the drift term of $\frac{dC}{C}$ must be zero:

$$q + \mu_s - \mu_y + \sigma_y^2 - \sigma_s\sigma_y\rho = 0$$

Solving for μ_s:

$$\mu_s = \mu_y - \sigma_y^2 + \sigma_s \sigma_y \rho - q$$

From Q1.1, we know that $\mu_y = r + \sigma_y^2$, so we substitute:

$$\mu_s = (r + \sigma_y^2) - \sigma_y^2 + \sigma_s \sigma_y \rho - q = r + \sigma_s \sigma_y \rho - q$$

Conclusion: The drift of $\frac{dS}{S}$ under the physical measure is:

$$\boxed{\frac{dS}{S} = (r + \sigma_s \sigma_y \rho - q)\, dt + \sigma_s\, dB_s}$$

Q1.3

We showed in Question 1.2 that the drift of the process $\frac{dS}{S}$ satisfies:

$$\mu_s = \mu_y - \sigma_y^2 + \sigma_s \sigma_y \rho - q$$

From Question 1.1, when Y is the numéraire, we have:

$$\mu_y = r + \sigma_y^2 \quad \Rightarrow \quad \mu_s = r + \sigma_s \sigma_y \rho - q$$

Case 1: $Y = R(t)$, the risk-free asset

In this case, $\sigma_y = 0$ and $\mu_y = r$, so the formula becomes:

$$\mu_s = r - q \quad \Rightarrow \quad \boxed{\frac{dS}{S} = (r - q)\, dt + \sigma_s\, dB_s}$$

Case 2: $Y = S(t)$, the risky stock itself

In this case, $Y = S$, so $\mu_y = \mu_s$, $\sigma_y = \sigma_s$, and $\rho = 1$. Substituting into the general formula:

$$\mu_s = \mu_s - \sigma_s^2 + \sigma_s^2 \cdot 1 - q = \mu_s - q \quad \Rightarrow \quad \mu_s = r - q + \sigma_s^2$$

Therefore:

$$\boxed{\frac{dS}{S} = (r - q + \sigma_s^2)\, dt + \sigma_s\, dB_s}$$

This confirms the result stated in class: when the risky stock is used as the numéraire, the drift of the stock process includes the adjustment $+\sigma_s^2$, and no correlation term appears, because both numerator and denominator are driven by the same Brownian motion.

Q2. Option valuation using the CAPM

- In the original Black and Scholes (1973) paper, there are *two* derivations of the option-pricing formula. One approach relies on constructing an option position that is delta-hedged with a stock, and because this position has zero risk, it is argued that it must earn just the risk-free rate of return; this argument is then used to derive the fundamental partial differential equation.
- The second approach for deriving the value of options relies on the Capital Asset Pricing Model (CAPM), and this question asks you to use that approach to derive the value of a call option. Recall that the CAPM states that the relation between the risk and return of a security is a linear function of its beta (β), where the β denotes the covariance between the return on the asset and the return on the market portfolio, divided by the variance of the return on the market portfolio.
- Assume the setting of a standard Black-Scholes economy with a risk-free bank account with price $R(t)$ that earns a constant risk-free rate r, and stock price $S(t)$ that follows a Geometric Brownian Motion with constant volatility σ_S and constant drift μ_S:

$$\frac{dS(t)}{S(t)} = \mu_S \, dt + \sigma_S \, dB_S(t).$$

- Finally, assume that the *return* on the market portfolio is given by:

$$dR_M(t) = \mu_M \, dt + \sigma_M \, dB_M(t),$$

  where μ_M and σ_M are constants, and $dB_S(t) \times dB_M(t) = \rho_{MS}$.
- Now, consider a call option with strike price K and maturity T. Denote the price today of this option by $C(S_t, t)$, where S_t is the price of the stock on the current date t.

- Let β_{MC} denote the beta of the call option with respect to the return on the market, and let β_{MS} denote the beta of the stock underlying this call option.

Please answer the following questions,

2.1 Write down (a) the instantaneous expected return on the call option according to the CAPM, and (b) use Itô's formula to identify the process for the return on the call option.

2.2 Find the covariance between the return on the call option and the return on the market portfolio. Then, use this result to identify the relation between the beta on the call, β_{MC}, and the beta on the stock, β_{MS}.

2.3 Use the results of the previous parts to derive the fundamental partial differential equation that the price of the call option must satisfy, which is:

$$r\,C = C_t + r\,C_S\,S + \frac{1}{2}\sigma_S^2\,C_{SS}\,S^2.$$

2.4 Explain which approach for deriving the option price is better: the approach based on constructing a risk-free hedge portfolio, or the approach based on the CAPM. Why?

Answer to Q2 starts here.

Q2.1

(a) Instantaneous expected return on call option according to the CAPM

According to the Capital Asset Pricing Model (CAPM), the expected return on any asset i is given by:

$$\mathbb{E}^{\mathbb{P}}\left[\frac{dR_i(t)}{R_i(t)}\right] = r + \beta_i\left(\mu_M - r\right),$$

where:

- r is the constant risk-free rate,
- μ_M is the expected return on the market portfolio,
- β_i is the beta of asset i with respect to the market, defined as:

$$\beta_i = \frac{\mathrm{Cov}\left(\frac{dR_i}{R_i}, \frac{dR_M}{R_M}\right)}{\mathrm{Var}\left(\frac{dR_M}{R_M}\right)}.$$

In our case, the asset in question is the European call option with value $C(t)$. Therefore, the CAPM implies that its expected return satisfies:

Small error:

(1/dt) $$\mathbb{E}^{\mathbb{P}}\left[\frac{dC(t)}{C(t)}\right] = r + \beta_{MC}\left(\mu_M - r\right),$$

where β_{MC} denotes the beta of the call option with respect to the return on the market portfolio, defined as:

$$\beta_{MC} = \frac{\mathrm{Cov}\left(\frac{dC(t)}{C(t)}, \frac{dR_M(t)}{R_M(t)}\right)}{\mathrm{Var}\left(\frac{dR_M(t)}{R_M(t)}\right)}.$$

This gives us the expression for the instantaneous expected return of the call option according to the CAPM, under the physical probability measure.

(b) Return Process from Itô's Formula

We are given that the stock price $S(t)$ follows the process:

$$\frac{dS(t)}{S(t)} = \mu_S \, dt + \sigma_S \, dB_S(t),$$

and that the call option has value $C(S,t)$. Applying Itô's lemma to $C(S,t)$, we obtain:

$$\begin{aligned} dC(t) &= C_t \, dt + C_S \, dS + \frac{1}{2} C_{SS} \, (dS)^2 \\ &= \left(C_t + \mu_S S C_S + \frac{1}{2} \sigma_S^2 S^2 C_{SS} \right) dt + \sigma_S S C_S \, dB_S(t) \end{aligned}$$

With $C_t = \frac{\partial C}{\partial t}$, $C_S = \frac{\partial C}{\partial S}$, and $C_{SS} = \frac{\partial^2 C}{\partial S^2}$. To obtain the return process, we divide both sides by $C(t)$:

$$\frac{dC}{C} = \left(\frac{C_t}{C} + \mu_S S \frac{C_S}{C} + \frac{1}{2} \sigma_S^2 S^2 \frac{C_{SS}}{C} \right) dt + \sigma_S S \frac{C_S}{C} \, dB_S(t)$$

Q2.2

The return on the call option is given by Itô's lemma:

$$\frac{dC}{C} = \left(\frac{C_t}{C} + \mu_S S \frac{C_S}{C} + \frac{1}{2} \sigma_S^2 S^2 \frac{C_{SS}}{C} \right) dt + \sigma_S S \frac{C_S}{C} \, dB_S(t)$$

The return on the market portfolio is:

$$\frac{dR_M}{R_M} = \mu_M \, dt + \sigma_M \, dB_M(t)$$

The Brownian motions satisfy:

$$dB_S(t) \, dB_M(t) = \rho_{MS} \, dt$$

Then, the instantaneous covariance between the two return processes is:

$$\begin{aligned}\operatorname{Cov}\left(\frac{dC}{C}, \frac{dR_M}{R_M}\right) &= \operatorname{Cov}\left(\sigma_S S \frac{C_S}{C}\, dB_S, \sigma_M\, dB_M\right) \\ &= \sigma_S S \frac{C_S}{C} \cdot \sigma_M \cdot \rho_{MS}\, dt\end{aligned}$$

Therefore, the beta of the call with respect to the market is:

$$\beta_{MC} = \frac{\operatorname{Cov}\left(\frac{dC}{C}, \frac{dR_M}{R_M}\right)}{\operatorname{Var}\left(\frac{dR_M}{R_M}\right)} = \frac{\sigma_S S \frac{C_S}{C} \cdot \sigma_M \cdot \rho_{MS}}{\sigma_M^2} = \frac{SC_S}{C} \cdot \rho_{MS} \cdot \frac{\sigma_S}{\sigma_M}$$

Similarly, the beta of the stock is:

$$\beta_{MS} = \frac{\operatorname{Cov}\left(\frac{dS}{S}, \frac{dR_M}{R_M}\right)}{\operatorname{Var}\left(\frac{dR_M}{R_M}\right)} = \frac{\sigma_S \cdot \sigma_M \cdot \rho_{MS}}{\sigma_M^2} = \rho_{MS} \cdot \frac{\sigma_S}{\sigma_M}$$

Combining the two expressions:

$$\boxed{\beta_{MC} = \frac{SC_S}{C} \cdot \beta_{MS}}$$

This relation expresses the beta of the option in terms of the beta of the underlying stock and the option delta.

Q2.3

From Question 2.1, we know that the instantaneous return on the call option satisfies:

$$\frac{dC}{C} = \left(\frac{C_t}{C} + \mu_S S \frac{C_S}{C} + \frac{1}{2}\sigma_S^2 S^2 \frac{C_{SS}}{C}\right) dt + \sigma_S S \frac{C_S}{C}\, dB_S(t)$$

Taking the expectation under the physical (actual) measure gives the drift of the call return:

$$\mathbb{E}^{\mathbb{P}}\left[\frac{dC}{C}\right] = \frac{C_t}{C} + \mu_S S\frac{C_S}{C} + \frac{1}{2}\sigma_S^2 S^2\frac{C_{SS}}{C}$$

From the CAPM (Question 2.1a), we also know:

$$\mathbb{E}^{\mathbb{P}}\left[\frac{dC}{C}\right] = r + \beta_{MC}(\mu_M - r)$$

From Question 2.2, we derived the relationship:

$$\beta_{MC} = \frac{SC_S}{C}\cdot\beta_{MS} \quad \text{and} \quad \beta_{MS} = \frac{\sigma_S\rho_{MS}}{\sigma_M}$$

Also, since the CAPM for the stock gives:

$$\mu_S = r + \beta_{MS}(\mu_M - r)$$

we substitute this into the Itô drift:

$$\mathbb{E}^{\mathbb{P}}\left[\frac{dC}{C}\right] = \frac{C_t}{C} + \left[r + \beta_{MS}(\mu_M - r)\right] S\frac{C_S}{C} + \frac{1}{2}\sigma_S^2 S^2\frac{C_{SS}}{C}$$

Now use the identity from Q2.2:

$$\beta_{MC} = \frac{SC_S}{C}\cdot\beta_{MS}$$

So:

$$r + \beta_{MC}(\mu_M - r) = r + \left(\frac{SC_S}{C}\cdot\beta_{MS}\right)(\mu_M - r) = r + \beta_{MS}(\mu_M - r)S\frac{C_S}{C}$$

Substitute into the CAPM equation:

$$\frac{C_t}{C} + \left[r + \beta_{MS}(\mu_M - r)\right] S\frac{C_S}{C} + \frac{1}{2}\sigma_S^2 S^2\frac{C_{SS}}{C} = r + \beta_{MS}(\mu_M - r)S\frac{C_S}{C}$$

Cancel matching terms:

$$\frac{C_t}{C} + rS\frac{C_S}{C} + \frac{1}{2}\sigma_S^2 S^2 \frac{C_{SS}}{C} = r$$

Multiply both sides by C:

$$C_t + rSC_S + \frac{1}{2}\sigma_S^2 S^2 C_{SS} = rC$$

Thus, the PDE that must be satisfied by the price of the call option is:

$$\boxed{rC = C_t + rSC_S + \frac{1}{2}\sigma_S^2 S^2 C_{SS}}$$

This matches the standard Black-Scholes PDE, but is now derived using the CAPM equilibrium condition rather than a no-arbitrage hedging argument.

Q2.4

The CAPM-based approach is *far less precise* than the no-arbitrage method. This is because the CAPM relies on *expectations* of future returns under the physical (real-world) probability measure. These expectations are inherently subjective and introduce uncertainty into the pricing formula. In contrast, the hedging-based approach does not involve any expectations: it is based purely on constructing a replicating portfolio whose value evolves deterministically when properly hedged. This makes the hedging approach fully objective, model-independent (within the no-arbitrage setting), and free from the estimation risk associated with expected returns.

Moreover, the hedging-based approach is more fundamental and preferable in modern derivative pricing theory because it does not require equilibrium assumptions about investor preferences or the distribution of returns—it only relies on the absence of arbitrage.

In addition, the CAPM approach depends on strong assumptions about market structure (e.g., mean-variance optimization), which are not needed in the no-arbitrage framework.

Q3. Options and hedging

- In class, we studied the Black-Scholes model of option pricing that assumes that the interest rate is constant. We also studied Merton's model of option pricing that does not require the assumption of a constant interest rate; the Merton model also allows for volatility that is changing over time, as long as the changes are deterministic (non-stochastic).
- Suppose the price $P^S(t)$ of a non-dividend paying stock has a constant volatility σ_S.
- Let the price of a discount bond be denoted by $P^D(t,T)$.
- Let the constant $\kappa > 0$ denote the rate of mean reversion of the short rate process and the constant $\phi > 0$ the (absolute) volatility of the short rate process.
- Assume that that volatility at date t of a discount bond maturing at $T > t$ is

  $$\sigma_D(t) = \frac{\phi}{\kappa}\big(1 - e^{-\kappa(T-t)}\big),$$

  and the return on the stock and bond have constant correlation ρ.

Please answer the following questions,

3.1 Find the volatility at t of $\frac{P^S(t)}{P^D(t,T)}$; that is, the volatility term of the process

$$\frac{d\left(\frac{P^S(t)}{P^D(t,T)}\right)}{\frac{P^S(t)}{P^D(t,T)}}.$$

Denote this volatility of $\frac{P^S(t)}{P^D(t,T)}$ by $\boldsymbol{\sigma}(t)$.

3.2 Define

$$\boldsymbol{\sigma}_{avg} = \sqrt{\frac{1}{T}\int_0^T \left[\boldsymbol{\sigma}(t)\right]^2 \, dt}.$$

Show that

$$\boldsymbol{\sigma}_{avg}^2 = \sigma_S^2 + \frac{1}{\kappa^2}\left[\phi^2 - 2\kappa\rho\sigma_S\phi - \left(2\phi^2 - 2\kappa\rho\sigma_S\phi\right)\left(\frac{1-e^{-\kappa T}}{\kappa T}\right) + \phi^2\left(\frac{1-e^{-2\kappa T}}{2\,\kappa T}\right)\right].$$

3.3 For small T, use l'Hôpital's rule to show that $\boldsymbol{\sigma}_{avg} \approx \sigma_S$.

3.4 For large T, show that $\boldsymbol{\sigma}_{avg} > \sigma_S$ if ρ is sufficiently small.

Answer to Q3 starts here.

Q3.1

Let us define the process:

$$Z(t) = \frac{P^S(t)}{P^D(t,T)}$$

and compute the volatility of $Z(t)$, that is, the coefficient of the Brownian motion in the stochastic differential of $\frac{dZ(t)}{Z(t)}$.

We assume the following dynamics under the physical measure:

$$\frac{dP^S(t)}{P^S(t)} = \mu_S\,dt + \sigma_S\,dW_S(t), \qquad \frac{dP^D(t,T)}{P^D(t,T)} = \mu_D(t)\,dt + \sigma_D(t)\,dW_D(t),$$

with correlation:

$$\mathbb{E}[dW_S(t)\,dW_D(t)] = \rho\,dt.$$

To derive the dynamics of the ratio, we apply Itô's formula to $Z(t) = \frac{P^S(t)}{P^D(t,T)}$, which gives:

$$\frac{dZ(t)}{Z(t)} = \frac{dP^S(t)}{P^S(t)} - \frac{dP^D(t,T)}{P^D(t,T)} + \left(\frac{dP^D(t,T)}{P^D(t,T)}\right)^2 - \frac{dP^S(t)}{P^S(t)} \cdot \frac{dP^D(t,T)}{P^D(t,T)}$$

Substituting the dynamics into the expression:

$$\begin{aligned}\frac{dZ(t)}{Z(t)} &= (\mu_S\,dt + \sigma_S\,dW_S(t)) - (\mu_D(t)\,dt + \sigma_D(t)\,dW_D(t)) + \sigma_D^2(t)\,dt - \rho\sigma_S\sigma_D(t)\,dt \\ &= \big(\mu_S - \mu_D(t) + \sigma_D^2(t) - \rho\sigma_S\sigma_D(t)\big)\,dt + (\sigma_S\,dW_S(t) - \sigma_D(t)\,dW_D(t))\end{aligned}$$

The volatility of $Z(t)$ corresponds to the diffusion term:

$$\sigma_S\,dW_S(t) - \sigma_D(t)\,dW_D(t)$$

The instantaneous variance of this term is:

$$\text{Var} = \sigma_S^2 + \sigma_D^2(t) - 2\rho\sigma_S\sigma_D(t)$$

Hence, the volatility of the process $Z(t) = \frac{P^S(t)}{P^D(t,T)}$ is:

$$\boxed{\boldsymbol{\sigma}(t) = \sqrt{\sigma_S^2 + \sigma_D^2(t) - 2\rho\sigma_S\sigma_D(t)}}$$

Q3.2

We are given the instantaneous volatility:

$$\boldsymbol{\sigma}^2(t) = \sigma_S^2 + \sigma_D^2(t) - 2\rho\sigma_S\sigma_D(t)$$

with:

$$\sigma_D(t) = \frac{\phi}{\kappa}\left(1 - e^{-\kappa(T-t)}\right)$$

Then:

$$\begin{aligned}\boldsymbol{\sigma}^2(t) &= \sigma_S^2 + \left(\frac{\phi}{\kappa}(1 - e^{-\kappa(T-t)})\right)^2 - 2\rho\sigma_S \cdot \frac{\phi}{\kappa}(1 - e^{-\kappa(T-t)}) \\ &= \sigma_S^2 + \frac{\phi^2}{\kappa^2}\left(1 - 2e^{-\kappa(T-t)} + e^{-2\kappa(T-t)}\right) - \frac{2\rho\sigma_S\phi}{\kappa}(1 - e^{-\kappa(T-t)})\end{aligned}$$

The average volatility over $[0, T]$ is:

$$\sigma_{\text{avg}}^2 = \frac{1}{T}\int_0^T \boldsymbol{\sigma}^2(t)\, dt$$

We integrate each term:

- $\int_0^T 1\, dt = T$
- $\int_0^T e^{-\kappa(T-t)}\, dt = \frac{1-e^{-\kappa T}}{\kappa}$
- $\int_0^T e^{-2\kappa(T-t)}\, dt = \frac{1-e^{-2\kappa T}}{2\kappa}$

Putting everything together:

$$\sigma_{\text{avg}}^2 = \sigma_S^2 + \frac{1}{\kappa^2}\left[\phi^2 - 2\kappa\rho\sigma_S\phi - \left(2\phi^2 - 2\kappa\rho\sigma_S\phi\right)\left(\frac{1 - e^{-\kappa T}}{\kappa T}\right) + \phi^2\left(\frac{1 - e^{-2\kappa T}}{2\kappa T}\right)\right]$$

Q3.3

We start from the expression for σ^2_{avg}:

$$\sigma^2_{\text{avg}} = \sigma^2_S + \frac{1}{\kappa^2}\left[\phi^2 - 2\kappa\rho\sigma_S\phi - (2\phi^2 - 2\kappa\rho\sigma_S\phi)\frac{1-e^{-\kappa T}}{\kappa T} + \phi^2\frac{1-e^{-2\kappa T}}{2\kappa T}\right]$$

For small T, we apply limits explicitly using L'Hôpital's rule:

$$\lim_{T\to 0}\frac{1-e^{-\kappa T}}{\kappa T} = \lim_{T\to 0}\frac{\kappa e^{-\kappa T}}{\kappa} = 1, \quad \lim_{T\to 0}\frac{1-e^{-2\kappa T}}{2\kappa T} = \lim_{T\to 0}\frac{2\kappa e^{-2\kappa T}}{2\kappa} = 1.$$

Thus, as $T \to 0$:

$$\sigma^2_{\text{avg}} \to \sigma^2_S + \frac{1}{\kappa^2}\left[\phi^2 - 2\kappa\rho\sigma_S\phi - (2\phi^2 - 2\kappa\rho\sigma_S\phi) + \phi^2\right] = \sigma^2_S.$$

Hence:

$$\lim_{T\to 0}\sigma_{\text{avg}} = \sigma_S,$$

confirming that as maturity approaches zero, average volatility converges exactly to the instantaneous volatility of the stock.

Q3.4

For large maturities ($T \to \infty$), we have:

$$\frac{1-e^{-\kappa T}}{\kappa T} \to 0, \quad \frac{1-e^{-2\kappa T}}{2\kappa T} \to 0.$$

Thus, the average volatility simplifies to:

$$\lim_{T\to\infty} \sigma_{\text{avg}}^2 = \sigma_S^2 + \frac{1}{\kappa^2}\left(\phi^2 - 2\kappa\rho\sigma_S\phi\right).$$

To analyze when $\sigma_{\text{avg}} > \sigma_S$, we require:

$$\sigma_{\text{avg}}^2 - \sigma_S^2 = \frac{1}{\kappa^2}(\phi^2 - 2\kappa\rho\sigma_S\phi) > 0.$$

Assuming $\phi, \kappa, \sigma_S > 0$, we obtain the inequality condition explicitly:

$$\rho < \frac{\phi}{2\kappa\sigma_S}.$$

This shows that for sufficiently small correlation ρ, the long-run average volatility exceeds the instantaneous volatility of the stock, σ_S.

Q4. Option valuation and hedging

- We now study the problem of an option dealer who has sold a particular option whose payoff depends on the prices of two stocks and the dealer wishes to hedge her position.
- Suppose that the price of the first stock, S_1 follows the process

$$\frac{dS_1(t)}{S_1(t)} = \mu_1 dt + \sigma_1 dB_1(t),$$

  and that the price of the second stock, S_2 follows the process

$$\frac{dS_2(t)}{S_2(t)} = \mu_2 dt + \sigma_2 dB_2(t),$$

  where μ_1, μ_2, σ_1, and σ_2 are constants, $dB_1(t)$ and $dB_2(t)$ are Brownian motions, with the correlation ρ between the two Brownian motions also being constant: $dB_1(t) \times dB_2(t) = \rho\, dt$.
- Let $D(t)$ denote the price of a bank-deposit that earns the constant risk-free rate of return:

$$\frac{dD(t)}{D(t)} = r\, dt.$$

- Let $C(t)$ denote the value at t of a European option written on the difference between the prices of these two stocks. Specifically, at the maturity date T of the option, the value of the option is

$$C(T) = \max\Big(0, S_1(T) - S_2(T)\Big),$$

Please answer the following questions,

4.1 Using Itô's formula, derive the process followed by the value of this option.

4.2 Suppose that you are an option dealer who has just sold (written) one of these options for a customer. You now wish to form a hedge portfolio composed of your unit short position in the option and positions in the two stocks. Let $H(t)$ denote the date t value of this hedge portfolio. Let $n_1(t)$ and $n_2(t)$ denote the number of shares in each of these two stocks at t. Write down the process for $H(t)$.

4.3 Identify the number of shares to hold in each of the two stocks so that the hedge portfolio $H(t)$ is riskless.

4.4 In the absence of arbitrage, derive the partial differential equation that must be satisfied by the option value $C(t)$. Does the risk-free rate appear in this differential equation? Explain why or why not?

Answer to Q4 starts here.

Q4.1

Let $C(t) = C(S_1(t), S_2(t), t)$ be the value of the option at time t, which is a function of both S_1 and S_2, and of time. Applying Itô's lemma to the function $C(t, S_1, S_2)$, we obtain:

$$dC = C_t\,dt + C_{S_1}\,dS_1 + C_{S_2}\,dS_2 + \frac{1}{2}C_{S_1S_1}\,(dS_1)^2 + \frac{1}{2}C_{S_2S_2}\,(dS_2)^2 + C_{S_1S_2}\,dS_1\,dS_2$$

$$= C_t\,dt + C_{S_1}(\mu_1 S_1\,dt + \sigma_1 S_1\,dB_1) + C_{S_2}(\mu_2 S_2\,dt + \sigma_2 S_2\,dB_2)$$
$$+ \frac{1}{2}C_{S_1S_1}\sigma_1^2 S_1^2\,dt + \frac{1}{2}C_{S_2S_2}\sigma_2^2 S_2^2\,dt + C_{S_1S_2}\rho\sigma_1\sigma_2 S_1 S_2\,dt$$

$$= \left(C_t + \mu_1 S_1 C_{S_1} + \mu_2 S_2 C_{S_2} + \frac{1}{2}\sigma_1^2 S_1^2 C_{S_1S_1} + \frac{1}{2}\sigma_2^2 S_2^2 C_{S_2S_2} + \rho\sigma_1\sigma_2 S_1 S_2 C_{S_1S_2}\right) dt$$
$$+ \sigma_1 S_1 C_{S_1}\,dB_1 + \sigma_2 S_2 C_{S_2}\,dB_2.$$

Q4.2

Consider a hedge portfolio composed of a short position in one unit of the option and positions in the two stocks. Let $n_1(t)$ and $n_2(t)$ denote the number of shares held in stocks S_1 and S_2, respectively. Then, the value of the hedge portfolio is:

$$H(t) = -C(t) + n_1(t)S_1(t) + n_2(t)S_2(t)$$

To determine the dynamics of $H(t)$, we differentiate using Itô's lemma:

$$dH(t) = -dC(t) + n_1(t)\,dS_1(t) + n_2(t)\,dS_2(t)$$

Substituting into the expression for $dH(t)$ the dynamics of the assets, we obtain:

$$\begin{aligned}
dH &= -dC(t) + n_1(t)\,dS_1(t) + n_2(t)\,dS_2(t) \\
&= -\left[C_t + \mu_1 S_1 C_{S_1} + \mu_2 S_2 C_{S_2} + \frac{1}{2}\sigma_1^2 S_1^2 C_{S_1S_1} + \frac{1}{2}\sigma_2^2 S_2^2 C_{S_2S_2} + \rho\sigma_1\sigma_2 S_1 S_2 C_{S_1S_2}\right] dt \\
&\quad - \sigma_1 S_1 C_{S_1}\,dB_1 - \sigma_2 S_2 C_{S_2}\,dB_2 \\
&\quad + n_1(\mu_1 S_1\,dt + \sigma_1 S_1\,dB_1) + n_2(\mu_2 S_2\,dt + \sigma_2 S_2\,dB_2)
\end{aligned}$$

Grouping drift and diffusion terms, we have:

$$\begin{aligned}
dH = \Big[&- C_t - \mu_1 S_1 C_{S_1} - \mu_2 S_2 C_{S_2} - \frac{1}{2}\sigma_1^2 S_1^2 C_{S_1S_1} - \frac{1}{2}\sigma_2^2 S_2^2 C_{S_2S_2} - \rho\sigma_1\sigma_2 S_1 S_2 C_{S_1S_2} \\
&+ n_1\mu_1 S_1 + n_2\mu_2 S_2\Big] dt \\
&+ (n_1 - C_{S_1})\,\sigma_1 S_1\,dB_1 + (n_2 - C_{S_2})\,\sigma_2 S_2\,dB_2.
\end{aligned}$$

This is the process followed by the hedge portfolio $H(t)$.

Q4.3

To make the hedge portfolio locally riskless, we eliminate the stochastic components by setting the coefficients of dB_1 and dB_2 in $dH(t)$ to zero:

$$(n_1 - C_{S_1})\sigma_1 S_1 = 0 \quad \Rightarrow \quad n_1(t) = C_{S_1}(t)$$

$$(n_2 - C_{S_2})\sigma_2 S_2 = 0 \quad \Rightarrow \quad n_2(t) = C_{S_2}(t)$$

Thus, the portfolio becomes locally riskless by holding the option deltas in the underlying assets. At this stage, we have not imposed any condi-

tion on the return of the portfolio, nor included a risk-free asset. Therefore, the portfolio is only instantaneously riskless, not yet in equilibrium with the market.

Q4.4

To price the option under the absence of arbitrage, we construct a self-financing replicating portfolio:

$$V(t) = a_1(t)S_1(t)+a_2(t)S_2(t)+b(t)D(t), \quad \text{with} \quad D(t) = e^{rt},\ \frac{dD(t)}{D(t)} = rdt$$

We impose the replication condition:

$$V(t) = C(t), \quad \text{and} \quad dV(t) = dC(t)$$

Using Itô's formula as before:

$$\begin{aligned} dC(t) = &\left(C_t + \mu_1 S_1 C_{S_1} + \mu_2 S_2 C_{S_2} + \tfrac{1}{2}\sigma_1^2 S_1^2 C_{S_1S_1} + \tfrac{1}{2}\sigma_2^2 S_2^2 C_{S_2S_2} + \rho\sigma_1\sigma_2 S_1 S_2 C_{S_1S_2}\right) dt \\ &+ \sigma_1 S_1 C_{S_1}\, dB_1 + \sigma_2 S_2 C_{S_2}\, dB_2 \end{aligned}$$

For the portfolio:

$$dV(t) = a_1 dS_1 + a_2 dS_2 + b\, dD = a_1(\mu_1 S_1 dt + \sigma_1 S_1 dB_1) + a_2(\mu_2 S_2 dt + \sigma_2 S_2 dB_2) + brDdt$$

Matching the diffusion terms:

$$a_1 = C_{S_1}, \quad a_2 = C_{S_2}$$

Matching the drift terms:

$$\begin{aligned} &C_t + \tfrac{1}{2}\sigma_1^2 S_1^2 C_{S_1S_1} + \tfrac{1}{2}\sigma_2^2 S_2^2 C_{S_2S_2} + \rho\sigma_1\sigma_2 S_1 S_2 C_{S_1S_2} \\ &= brD + C_{S_1}\mu_1 S_1 + C_{S_2}\mu_2 S_2 - C_{S_1}\mu_1 S_1 - C_{S_2}\mu_2 S_2 \\ &= brD \end{aligned}$$

Since $C(t) = a_1 S_1 + a_2 S_2 + bD$, we get:

$$b = \frac{C(t) - C_{S_1} S_1 - C_{S_2} S_2}{D}$$

Substitute back:

$$C_t + \tfrac{1}{2}\sigma_1^2 S_1^2 C_{S_1 S_1} + \tfrac{1}{2}\sigma_2^2 S_2^2 C_{S_2 S_2} + \rho\sigma_1\sigma_2 S_1 S_2 C_{S_1 S_2} = r\left(C - C_{S_1} S_1 - C_{S_2} S_2\right)$$

Rearranging, we obtain the pricing PDE:

$$\boxed{C_t + \tfrac{1}{2}\sigma_1^2 S_1^2 C_{S_1 S_1} + \tfrac{1}{2}\sigma_2^2 S_2^2 C_{S_2 S_2} + \rho\sigma_1\sigma_2 S_1 S_2 C_{S_1 S_2} + r S_1 C_{S_1} + r S_2 C_{S_2} - rC = 0}$$

This PDE reflects the fact that the riskless portfolio must grow at the risk-free rate r in a no-arbitrage equilibrium. The risk-free rate r appears explicitly in the partial differential equation through the terms $rS_1 C_{S_1}$, $rS_2 C_{S_2}$, and $-rC$. This is because the replicating portfolio is constructed to be riskless and self-financing, and thus, under the no-arbitrage condition, it must grow at the risk-free rate. This requirement is imposed on the drift of the option value process, which leads directly to the presence of the risk-free rate in the pricing PDE.

This would not appear if you recognized that the option payoff is linear homogeneous.

Q5. Valuation of a forward-start option

- We wish to value a *forward-start option*. A forward-start option is an option for which the strike price is set equal to the stock price at some later date. In essence, it is an option that is issued at a later date with the strike set so that the option is at-the-money on the issue date. Often, such options are promised to corporate executives.
- Let us denote by T^{mat} the maturity date of the option, and by T^{set} the date on which the strike price of the option is set. Thus, we have that $T^{set} < T^{mat}$.
- Assume that the setting is of a standard Black-Scholes economy where there is a bond with price $R(t)$ that earns a constant risk-free rate r, and the stock price $S(t)$ follows a Geometric Brownian Motion with constant volatility σ and constant dividend yield q. That is:

$$\frac{dR}{R} = r\,dt,$$

$$\frac{dS}{S} = \mu\,dt + \sigma\,dB.$$

Please answer the following questions,

5.1 Write down the process for the price of the stock under the risk-neutral measure. Explain, in words, the logic underlying the drift and volatility of this process. Then, write the payoff to a forward-start call option on the maturity date T^{mat}.

5.2 Define an indicator variable x whose value is equal to 1 if the option is exercised, and is equal to 0 if the option is not exercised. Choose as your numeraire $V(t) = e^{q\,t} S(t)$. Find the value at $t = 0$ of the payoff at maturity of $H(T^{mat}) = x\, S(T^{mat})$.

5.3 Find the value at $t = 0$ of the payoff at maturity of $G(T^{mat}) = x\, S(T^{set})$. To value this, note that for $t < T^{set}$, $S(T^{set})$ is random, but once the strike price is fixed to be $S(T^{set})$ at $t = T^{set}$, then it is no longer random.

Hint: To value this payoff, choose as the numeraire the following *portfolio*, whose value we will denote as $Z(t)$:

i. Purchase $e^{-q\,T^{set}}$ shares of the stock at $t = 0$ and reinvest the dividends until date T^{set}, which will result in the ownership of one share worth $S(T^{set})$ at the date T^{set};

ii. At date T^{set}, sell the share and invest the proceeds in the risk-free asset until T^{mat}, whose value at $t = T^{mat}$ is: $S(T^{set})\, e^{r(T^{mat} - T^{set})}$.

5.4 Next, we need to calculate the probability of exercising the option under the two numeraires. So, find the probability of $S(T^{mat}) > S(T^{set})$ when $V(t)$ is used as the numeraire; that is, find $\text{prob}^V\,[S(T^{mat}) > S(T^{set})]$.

Similarly, find the probability of $S(T^{mat}) > S(T^{set})$ when $Z(t)$ is used as the numeraire; that is, find $\text{prob}^Z\,[S(T^{mat}) > S(T^{set})]$. *Hint:* Note that between T^{set} and T^{mat} the portfolio $Z(t)$ earns the risk-free rate.

5.5 Combining the answers to the questions above, please give the final expression for the value at $t = 0$ of a forward-start call option.

Then, provide a simple interpretation in plain English of the expression you have obtained for the value of the forward-start call option in the previous question.

Answer to Q5 starts here.

Q5.1

The stock price under the real-world measure $\mathbb{P}$ evolves as:

$$\frac{dS(t)}{S(t)} = \mu\, dt + \sigma\, dB^{\mathbb{P}}(t)$$

To determine the risk-neutral dynamics, we apply Girsanov's theorem. Let $W^{\mathbb{Q}}(t)$ be the Brownian motion under the risk-neutral measure $\mathbb{Q}$, defined by:

$$dW^{\mathbb{Q}}(t) = dB^{\mathbb{P}}(t) + \lambda\, dt$$

where the market price of risk λ is given by:

$$\lambda = \frac{\mu - (r - q)}{\sigma}$$

Substituting into the original SDE:

$$\begin{aligned}\frac{dS(t)}{S(t)} &= \mu\, dt + \sigma\, dB^{\mathbb{P}}(t) \\ &= \mu\, dt + \sigma\left(dW^{\mathbb{Q}}(t) - \lambda dt\right) \\ &= (\mu - \sigma\lambda)\, dt + \sigma\, dW^{\mathbb{Q}}(t)\end{aligned}$$

Substituting λ gives:

$$\frac{dS(t)}{S(t)} = (r - q)\, dt + \sigma\, dW^{\mathbb{Q}}(t)$$

Thus, under the risk-neutral measure $\mathbb{Q}$, the stock grows at rate $r - q$ and retains the same volatility.

Payoff at maturity of a forward-start call option: At time T^{mat}, the payoff of a forward-start call option (with strike set at T^{set} to equal $S(T^{\text{set}})$) is:

$$\max\left(S(T^{\text{mat}}) - S(T^{\text{set}}),\, 0\right)$$

Q5.2

We define the indicator variable x as:

$$x = \begin{cases} 1 & \text{if } S(T^{\text{mat}}) > S(T^{\text{set}}) \\ & \\ 0 & \text{otherwise} \end{cases} = \mathbf{1}_{\{S(T^{\text{mat}}) > S(T^{\text{set}})\}}$$

The payoff at maturity is:

$$H(T^{\text{mat}}) = x \cdot S(T^{\text{mat}}) = \mathbf{1}_{\{S(T^{\text{mat}}) > S(T^{\text{set}})\}} \cdot S(T^{\text{mat}})$$

We are asked to evaluate the value of this payoff at $t = 0$ using the numéraire $V(t) = e^{qt}S(t)$, which corresponds to the measure $\mathbb{Q}^V$ (the stock measure).

Using the change of numéraire approach, we write:

$$H(0) = S(0)e^{-qT^{\text{mat}}} \text{ prop}^{\mathbb{V}} \left(S(T^{\text{mat}}) > S(T^{\text{set}})\right)$$

This is the present value of receiving one share of the stock at time T^{mat}, conditional on the forward-start call option being in-the-money, computed under the stock numéraire.

Q5.3

We consider the payoff:

$$G(T^{\text{mat}}) = \mathbf{1}_{\{S(T^{\text{mat}}) > S(T^{\text{set}})\}} \cdot S(T^{\text{set}})$$

We use as numéraire the portfolio $Z(t)$, defined as:

- At $t = 0$: invest $e^{-qT^{\text{set}}}$ shares in the stock, accounting for continuous dividends,
- At T^{set}: own 1 share worth $S(T^{\text{set}})$, reinvest in risk-free asset,
- At T^{mat}: portfolio value is $Z(T^{\text{mat}}) = S(T^{\text{set}}) \cdot e^{r(T^{\text{mat}} - T^{\text{set}})}$.

We compute the price at $t = 0$ using change of numéraire:

$$G(0) = Z(0) \cdot \mathbb{E}^{\mathbb{Q}^Z} \left[\frac{G(T^{\text{mat}})}{Z(T^{\text{mat}})} \right] = Z(0) \text{prob}^{\mathbb{Z}} \left(S(T^{\text{mat}}) > S(T^{\text{set}}) \right)$$

Substituting:

$$Z(0) = e^{-qT^{\text{set}}} S(0), \quad \frac{G(T^{\text{mat}})}{Z(T^{\text{mat}})} = e^{-r(T^{\text{mat}} - T^{\text{set}})} \cdot \mathbf{1}_{\{S(T^{\text{mat}}) > S(T^{\text{set}})\}}$$

Thus, the price is:

$$G(0) = S(0) \cdot e^{-qT^{\text{set}}} \cdot e^{-r(T^{\text{mat}} - T^{\text{set}})} \text{prob}^{\mathbb{Z}} \left(S(T^{\text{mat}}) > S(T^{\text{set}}) \right)$$

Q5.4

We compute the probability that $S(T^{\text{mat}}) > S(T^{\text{set}})$ under two different numéraires.

Let $\tau = T^{\text{mat}} - T^{\text{set}}$. Then, the log return between the two dates is normally distributed:

$$\log \left(\frac{S(T^{\text{mat}})}{S(T^{\text{set}})} \right) \sim \mathcal{N}(\mu^* \tau, \sigma^2 \tau)$$

So, the general expression is:

$$\mathbb{P}^* \left[S(T^{\text{mat}}) > S(T^{\text{set}}) \right] = \Phi \left(\frac{\mu^* \sqrt{\tau}}{\sigma} \right)$$

(a) Under the numeraire $V(t) = e^{qt}S(t)$, the drift of $S(t)$ becomes $(r - q + \sigma^2)$ due to Itô's lemma applied to the dividend-adjusted numeraire, thus clearly giving:

$$\text{prob}^{\mathbb{V}}\left(S(T^{\text{mat}}) > S(T^{\text{set}})\right) = \Phi\left(\frac{(r - q + \sigma^2)\sqrt{\tau}}{\sigma}\right)$$

(b) Under the numéraire $Z(t)$**:** Under the measure $\mathbb{Q}^Z$, the portfolio $Z(t)$ earns the risk-free rate r, so the drift of $S(t)$ is r. Therefore:

$$\text{prob}^{\mathbb{Z}}\left(S(T^{\text{mat}}) > S(T^{\text{set}})\right) = \Phi\left(\frac{r\sqrt{\tau}}{\sigma}\right)$$

Q5.5

From Question 5.2:

$$H(0) = S(0) \cdot e^{-qT^{\text{mat}}}\text{prob}^{\mathbb{V}}\left(S(T^{\text{mat}}) > S(T^{\text{set}})\right)$$

From Question 5.3:

$$G(0) = S(0) \cdot e^{-qT^{\text{set}}} \cdot e^{-r(T^{\text{mat}} - T^{\text{set}})}\text{prob}^{\mathbb{Z}}\left(S(T^{\text{mat}}) > S(T^{\text{set}})\right)$$

Therefore, the value of the forward-start call option is:

$$\begin{aligned}\text{Call}(0) &= H(0) - G(0) \\ &= S(0)\Big[e^{-q\,T^{\text{mat}}}\text{prob}^{\mathbb{V}}\left(S(T^{\text{mat}}) > S(T^{\text{set}})\right) \\ &\qquad - e^{-q\,T^{\text{set}} - r\,(T^{\text{mat}} - T^{\text{set}})}\text{prob}^{\mathbb{Z}}\left(S(T^{\text{mat}}) > S(T^{\text{set}})\right)\Big]\end{aligned} \tag{1}$$

Economic Intuition: The forward-start call option allows the investor to lock in a right to buy the stock at a future unknown strike (set at $S(T^{\text{set}})$). Its value reflects two components:

- The value of receiving the stock if in-the-money at maturity (Q5.2),
- Minus the cost of paying the strike only if exercised (Q5.3).

Because the drift under $\text{prob}^{\mathbb{V}}$, includes an extra σ^2 compared to $\text{prob}^{\mathbb{Z}}$, the probability of exercise is higher under $\text{prob}^{\mathbb{V}}$. Thus, the forward-start call option has positive value, and this difference in probability under the two numéraires captures the option's risk-adjusted value.